Math 526, F06
Exam 2

Name: Answer
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	15	
2	20	
3	15	
4	20	
5	15	
6	15	
Total	100	

Good Luck !

Problem 1. (15 points) Let $\mathbf{x} = \begin{bmatrix} 1 \\ -6 \\ 4 \\ 2 \end{bmatrix}$, $\mathbf{A} = \begin{bmatrix} 2 & 4 & 0 \\ -1 & -2 & 5 \\ 2 & 4 & -5 \end{bmatrix}$, and $\mathbf{B} = \begin{bmatrix} -2 & 3 \\ 2 & -1 \end{bmatrix}$.

a) Compute $||\mathbf{x}||_1$, $||\mathbf{x}||_2$, and $||\mathbf{x}||_\infty$.

$$||x||_1 = |1| + |-6| + |4| + |2| = 13$$

$$||x||_2 = \sqrt{1^2 + (-6)^2 + 4^2 + 2^2} = \sqrt{57}$$

$$||x||_\infty = \max(|1|, |-6|, |4|, |2|) = 6$$

b) Compute $||\mathbf{A}||_1$ and $||\mathbf{A}||_\infty$.

$$||A||_1 = \max(2+1+2, 4+2+4, 0+5+5) = 10$$

$$||A||_\infty = \max(2+4+0, \quad +2+5, 2+4+5) = 11$$

d) Compute $\kappa_\infty(\mathbf{B})$.

$$B^{-1} = \frac{1}{2-2\cdot 3}\begin{bmatrix} -1 & 3 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} \\ \frac{2}{4} & \frac{2}{4} \end{bmatrix}$$

$$\kappa_\infty(B) = ||B||_\infty \cdot ||B^{-1}||_\infty$$

$$= 5 \cdot 1$$

$$= 5$$

Problem 2. (20 points) Consider the system $\mathbf{Ax} = \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 5 & -2 & 1 \\ 1 & 4 & 1 \\ -2 & 1 & -5 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 5 \\ 3 \\ 12 \end{bmatrix}$$

a) Write down the corresponding iterative formula for the **Gauss-Seidel method**, then compute the approximate solutions $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ of the system by the Gauss-Seidel method with the initial guess $\mathbf{x}_0 = \mathbf{0}$.

$$\underbrace{\begin{bmatrix} 5 & 0 & 0 \\ 1 & 4 & 0 \\ -2 & 1 & -5 \end{bmatrix}}_{S} \begin{bmatrix} x_1^{new} \\ x_2^{new} \\ x_3^{new} \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ 12 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 2 & -1 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}}_{T} \begin{bmatrix} x_1^{old} \\ x_2^{old} \\ x_3^{old} \end{bmatrix}$$

$$\begin{aligned} 5x_1^{new} &= 5 + 2x_2^{old} - x_3^{old} \\ 4x_2^{new} &= 3 - x_3^{old} - x_1^{new} \\ -5x_3^{new} &= 12 + 2x_1^{new} - x_2^{new} \end{aligned} \Rightarrow \begin{cases} x_1^{new} = \dfrac{5 + 2x_2^{old} - x_3^{old}}{5} \\ x_2^{new} = \dfrac{3 - x_3^{old} - x_1^{new}}{4} \\ x_3^{new} = \dfrac{12 - 2x_1^{new} \; x_2^{new}}{5} \end{cases}$$

$$x_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 5+2(0)-0 \\ \\ \\ \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{3-0-1}{4} \\ \\ \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{2} \\ \frac{12+2(1)-(\frac{1}{2})}{-5} \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{2} \\ -\frac{27}{10} \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 1.74 \\ 0.99 \\ -2.898 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} 1.9756 \\ 0.9806 \\ -2.99412 \end{bmatrix}$$

b) Compute the error measure $||\mathbf{x}_3 - \mathbf{x}||_\infty$ where the exact solution $\mathbf{x} = [2 \;\; 1 \;\; -3]^t$.

$$||x_3 - x||_\infty = \left|\left| \begin{pmatrix} 1.9756 - 2 \\ 0.9806 - 1 \\ -2.99412 + 3 \end{pmatrix} \right|\right|_\infty = 0.0244$$

Problem 3. (15 points) Consider the following equation

$$\begin{bmatrix} 6 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 6 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 6 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 6 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 6 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 6 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 6 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

The **Jacobi method** is applied to solve this system.
a) Compute the convergence rate $\|\mathbf{S}^{-1}\mathbf{T}\|_\infty$.

$$S^{-1}T = \begin{bmatrix} 0 & -\frac{1}{6} & 0 & 0 & -\frac{1}{6} & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{6} & 0 & 0 & -\frac{1}{6} & 0 & 0 \\ & & & \cdots & & & & \\ 0 & 0 & 0 & -\frac{1}{6} & 0 & 0 & -\frac{1}{6} & 0 \end{bmatrix}$$

$$\Rightarrow \|S^{-1}T\| = \frac{2}{6} = \frac{1}{3}$$

b) How many steps are needed to reduce the initial solution error (in the $\|\cdot\|_\infty$ norm) by a factor of 10^{-6}?

$$\left(\frac{1}{3}\right)^k \leq 10^{-6}$$

$$k \geq \frac{\ln(10^{-6})}{\ln(\frac{1}{3})} \approx 12.575$$

Since k must be an integer, we need 13 steps.

Problem 4. (20 points)

a) How many significant didgits does the approximation $x' = 0.02235$ of $x = 0.02276$ have?

$$x = 0.02276 = 2.276 \times 10^{-2}$$

$$|x - x'| = 0.00041 = 0.041 \times 10^{-2}$$

$$0.5 \times 10^{-2} \overset{k}{\leq} 0.041 \leq 0.5 \times 10^{-1}$$

$$\Rightarrow (0.5 \times 10^{-2}) \times 10^{-2} \leq |x - x'| \leq (0.5 \times 10^{-1}) \times 10^{-2}$$

So it has 2 significant digits.

b) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & -1 & 4 & -9 \\ 0 & \boxed{1} & 0 & 0 \\ 3 & 2 & 7 & -4 \end{bmatrix}$. Compute $det(\mathbf{A})$. Is $\mathbf{A}$ singular?

$$\det A = (-1) \cdot 1 \cdot \det \begin{bmatrix} 1 & 3 & 4 \\ 1 & 4 & -9 \\ 3 & 7 & -4 \end{bmatrix}$$

$$= -\det \begin{bmatrix} 4 & -9 \\ 7 & -4 \end{bmatrix} + 3 \cdot \det \begin{bmatrix} 1 & -9 \\ 3 & -4 \end{bmatrix} - 4 \det \begin{bmatrix} 1 & 4 \\ 3 & 7 \end{bmatrix}$$

$$= -47 + 69 - 20$$

$$= 42$$

So A is nonsingular.

Problem 5. (15 points)

a) Let $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}$. Is the matrix $\mathbf{I}_{4\times 4} - \mathbf{u}\mathbf{v}^t$ nonsingular? Explain.

$$V^t u = [1\ -1\ 2\ 1]\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = 1+1+2-1 = 3 \neq 1$$

so $1 - V^t u \neq 0$, i.e., $I_{4\times4} - uV^t$ is nonsingular

by the sherman-Morrison Formula.

b) Is it correct that $det \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} = det \begin{bmatrix} d & b & a & c \\ -h & -f & -e & -g \\ -l & -j & -i & -k \\ p & n & m & o \end{bmatrix}$? Explain. (Use the properties of determinats)

It is correct.

$$det\begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} = -det\begin{bmatrix} c & b & a & d \\ g & f & e & h \\ k & j & i & l \\ o & n & m & p \end{bmatrix} = det\begin{bmatrix} d & b & a & c \\ h & f & e & g \\ l & j & i & k \\ p & n & m & o \end{bmatrix}$$ Property 3.

$$= -det\begin{bmatrix} d & b & a & c \\ -h & -f & -e & -g \\ l & j & i & k \\ p & n & m & o \end{bmatrix} = det\begin{bmatrix} d & b & a & c \\ -h & -f & -e & -g \\ -l & -j & -i & -k \\ p & n & m & o \end{bmatrix}$$ Property 1

Problem 6. (15 points) Ture or False (no explaination needed).

a) Let $\mathbf{x}$ be a column 4-vector, then $||\mathbf{x}||_1 \leq 2||\mathbf{x}||_2$.

True

b) Let $\mathbf{A}$ be a 3×3 matrix, then $\kappa_1(\mathbf{A}) \leq 3\kappa_\infty(\mathbf{A})$.

False

c) Let $\mathbf{A} = \begin{bmatrix} -8 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then $\kappa_2(\mathbf{A}) = 3$.

False

d) The Gauss-Seidel method doesn't converge for the system $\begin{bmatrix} 1 & -0.4 \\ 2 & 1 \end{bmatrix} \mathbf{x} = \mathbf{b}$.

False

e) $\mathbf{A}$ and $\mathbf{B}$ are two $n \times n$ matrix, we have $det(\mathbf{AB}) = det(\mathbf{A}^t\mathbf{B}^t)$.

True